Equity Models for Quants

Pricing, Hedging, Smile & PnL Across Equity Products
Why Desks Choose Black-Scholes, Heston, Local Vol, LSV and When They Break

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Who This Is For

1. Equity Quants: You're calibrating models daily but need to explain *why* you chose Heston over Local Vol or LSV over Black-Scholes. This gives you the PnL language.
2. Derivatives Traders: You quote vanillas, barriers, cliquets, autocallables. You need to know which model risk you're warehousing and how to hedge it.
3. Risk/XVA Quants: You compute CVA, PFE, capital. Equity models drive everything (spot, vol surfaces, correlation, dividends).
If you're on an equity derivatives desk, this is your playbook. If you're interviewing for one, this is your cheat sheet.

Prerequisites

Must know:

- Stochastic calculus: SDEs, Itô's Lemma
- Black-Scholes intuition (we build on it)
- Basic equity derivatives: vanillas, barriers, variance swaps
- Greeks: Delta, Gamma, Vega in equity context

Not needed: Heavy PDE theory, measure theory. We focus on modeling **logic** and **consequences**, not proofs.

How to Use This Book

First pass (Weeks 1-2): Read Modules 1-4 cover-to-cover. Internalize delta bleed, gamma, and why Black-Scholes fails for exotics.
Second pass (Weeks 3-4): Attack Modules 5-7. Understand Heston calibration, Local Vol dynamics, LSV hedging.
Before interview (48 hours): Reread Module 8 interview Q&A. Memorize the model selection decision tree.
On the desk: Keep Module 7 (PnL Attribution) open during risk meetings. Use Module 6 (LSV) when pricing autocallables.

Pro Tips from the Desk

1. "Model = Hedge": Always ask: "What can I hedge with this model?" If you can't hedge it, you shouldn't price with it.
2. "Delta Bleeds in Vol": In equity, your delta hedge decays daily unless you model vol dynamics correctly. Heston > Black-Scholes.
3. "Skew is Crash Insurance": Negative skew isn't arbitrageit's the market pricing gap risk. Don't flatten it away.
4. "Local Vol is a Snapshot": It fits today's surface but lies about tomorrow. Use it for barriers, not for forward-starting exotics.
5. "LSV = The Compromise": Most desks use Local-Stochastic Vol because it fits and kind of works. But it's slow and parameter-heavy.

For equity forward with discrete dividends D_i at times t_i:

$$F(0,T) = \left(S_0 - \sum_{i=1}^{n} D_i e^{-rt_i} \right) e^{rT}$$

For continuous dividend yield q:

$$F(0,T) = S_0 e^{(r-q)T}$$

Example: SPX at 4500, $r = 4\%$, $q = 1.5\%$, $T = 1\text{Y}$

$$F = 4500 \times e^{(0.04-0.015)\times 1} = 4500 \times 1.0253 = 4613.85$$

Error from ignoring dividends: $\approx$ 2.5% on forward $\rightarrow$ 10-15% delta error.

1.8 Quick Interview Cheat Sheet Module 1

Module 1: Equities Trend, Rates Revert

Core Concept: Equity risk = delta/gamma/vega/skew. Dividends are critical.
Must-Know Formula:

$$F(0,T) = S_0 e^{(r-q)T}$$

Interview Triggers:

- "Why negative skew?" "Leverage effect + crash demand"
- "Wrong forward?" "Delta hedge bleeds"
- "Dividend risk?" "Discrete vs continuous matters"

Desk Wisdom:

- Always check dividend forecast vs market borrow rate
- Skew is insurance, not arbitrage

Common Trap: "Don't say 'dividends are continuous.' Most desks use discrete models."
Memory Aid: "Forward = Spot + Carry. Skew = Fear."
Interviewer Is Testing: Do you understand **carry** vs **dynamics**? Do you respect skew?

Interview Template

"For [PRODUCT], I'd use [MODEL] because..."
Step 1: Identify dominant risk (delta/vega/skew/barrier/forward)
Step 2: Match to model strength
Step 3: Acknowledge caveat/limitation
Step 4: Mention fallback/hedge strategy
Example: "For a 1Y autocallable, I'd use LSV because it captures both today's barrier condition AND forward skew dynamics. Caveat: it's slow, so I'd pre-compute the local vol surface. If time-critical, I'd use Heston with barrier adjustment."

8.2 Classic Equity Interview Questions

1. Why is equity skew structurally negative?

Question: Skew intuition.
Answer: "Three reasons: (1) Leverage effect: company becomes more volatile as equity falls spot-vol correlation $\rho < 0$. (2) Crash risk: market prices tail events (1987, 2008, 2020), bidding up OTM puts. (3) Supply/demand: investors buy puts for protection, dealers short skew persistent premium. It's not arbitrageit's insurance."

2. Why does local vol overprice barriers?

Question: Dynamics failure.
Answer: "Local vol assumes vol is deterministic function of spot. When spot approaches barrier, local vol predicts vol collapses (since ATM vol is lower than barrier vol). But in reality, vol often spikes at barriers due to hedging pressure. LSV or stochastic vol models capture this better. Local vol underprimes vol-of-vol misprices barrier by 3-5%."

3. When does Heston fail?

Question: Model limitations.
Answer: "Short-dated options (<1M): Heston can't fit steep skew without ξ blowing up (unstable). Extreme tails: Heston has thin tails vs reality (need jumps). Vol term structure: single κ forces all maturities to mean-revert at same speed. Fix: use Bates (add jumps) for short dates, or multi-factor Heston for term structure."

4. Why do delta hedges bleed?

Question: Vanna effect.
Answer: "Delta hedge assumes vol unchanged. When vol moves, delta shifts via vanna: $\partial\Delta/\partial\sigma$. If you're long an OTM put and vol spikes up, delta becomes more negative your short futures hedge is now too small you lose money. The bleed is vanna Œ vol move Œ spot move. Heston hedges this; BS doesn't."

5. How does dividend mis-modeling show up in PnL?

Question: Carry risk.
Answer: "Wrong forward price. If you model continuous div at 2% but market expects 4% (special dividend), your forward is too high by 2% Œ T. This makes calls overpriced and puts underpriced. Your delta hedge will be wrong: you'll be overhedged on calls (short too much delta) and underhedged on puts. PnL bleed appears as residual of $10k-50k/day depending on size. Fix: use discrete dividend model or adjust borrow rate q."